Rojin Roy

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Professional Summary

Embedded Firmware Engineer with 4+ years of experience writing embedded C for PIC32-based systems: communication protocols (UART, SPI, TCP/IP), interrupt-driven state machines, peripheral integration, and hardware-level debugging. Self-directed embedded Linux work including Linux kernel build from source, U-Boot bring-up on BeagleBone Black (AM335x), and PRU subsystem bring-up via the remoteproc framework. Looking for firmware or embedded Linux roles with an R&D focus where I can own and ship real features.

Experience

Unistring Tech Solutions Pvt. Ltd., Hyderabad
Application Engineer Oct 2025 – Present
Junior Application Engineer Mar 2022 – Sep 2025

- Wrote embedded C firmware for PIC32-based systems: UART and SPI communication, interrupt handling, timers, PWM, and state machine logic.
- Integrated peripheral interfaces at the firmware level, working directly with hardware registers and datasheets.
- Automated manual test procedures by scripting the communication protocol, replacing manual steps with a repeatable automated flow.
- Developed C-based test tooling using Win32 APIs.
- Diagnosed firmware and communication faults (UART, Ethernet/TCP-IP) using oscilloscopes, logic analyzers, Wireshark, and firmware logs — tracing failures to root cause in code.

Projects

Linux Kernel Build and Bring-up — BeagleBone Black (AM335x)

- Built U-Boot and the Linux kernel from source for BeagleBone Black using an ARM cross-compilation toolchain (`arm-linux-gnueabihf`).
- Understood and configured the AM335x boot sequence: ROM bootloader → MLO (first-stage, DRAM init) → U-Boot → kernel.
- Configured the kernel using `menuconfig`; analysed boot logs (`dmesg`) to verify driver and hardware initialisation.

PRU Subsystem Bring-up via remoteproc — BeagleBone Black

- Configured and built a kernel with `CONFIG_REMOTEPROC` to enable the remoteproc framework for coprocessor lifecycle management.
- Loaded PRU firmware into PRU instruction RAM at runtime via remoteproc and released the PRU from reset; verified execution through sysfs and kernel logs.

Technical Skills

Languages: C
Firmware: Interrupts, Timers, PWM, State Machines, Register-level peripheral programming
Protocols: UART, SPI, I^2C, TCP/IP, Ethernet
Embedded Linux: Kernel build (from source), U-Boot, ARM cross-compilation, remoteproc, sysfs, dmesg
Tools: Git, GCC, Make, Oscilloscope, Logic Analyzer, Wireshark
Platforms: PIC32, BeagleBone Black (AM335x)

Education

Bachelor of Technology (B.Tech)
Guru Nanak Institute of Technology, Hyderabad